Concurrency Control

Three Concurrency Problems

- In a multi-processing environment transactions can interfere with each other.
- Three concurrency problems can arise, that any DBMS must account for and avoid:
  - Lost Update
  - Uncommitted Dependency
  - Inconsistent Analysis

Transaction A	Time	Transaction B
–		–
–		–
RETRIEVE t	t1	–
–		–
–	t2	RETRIEVE t
–		–
UPDATE t	t3	–
–		–
–	t4	UPDATE t
–		–

Fig. 16.1 Transaction *A* loses an update at time *t4* .

The Lost Update Problem

A lost update occurs when a second transaction reads the state of the database prior to the first one writing a change, and then stomps on the first one's change with its own update.

Transaction A	Time	Transaction B
–		–
–		–
–	t1	UPDATE t
–		–
RETRIEVE t	t2	–
–		–
–	t3	ROLLBACK
–		

Fig. 16.2 Transaction *A* becomes dependent on an uncommitted change at time *t2*

The Uncommitted Dependency Problem

An uncommitted dependency occurs when a second transaction relies on a change which has not yet been committed, which is rolled back after the second transaction has begun.

The Inconsistent Analysis Problem

An inconsistent analysis occurs when totals are calculated during interleaved updates.

ACC 1	ACC 2	ACC 3
40	50	30

Transaction A	*Time*	*Transaction B*
–		–
–		–
RETRIEVE ACC 1 :	*t1*	–
sum = 40		–
–		–
RETRIEVE ACC 2 :	*t2*	–
sum = 90		–
–		–
–	*t3*	RETRIEVE ACC 3
–		–
–	*t4*	UPDATE ACC 3 :
–		30 ⟶ 20
–		–
–	*t5*	RETRIEVE ACC 1
–		–
–	*t6*	UPDATE ACC 1 :
–		40 ⟶ 50
–		–
–	*t7*	COMMIT
–		
RETRIEVE ACC 3 :	*t8*	
sum = 110, *not* 120		
–		

Fig. 16.4 Transaction *A* performs an inconsistent analysis

Conflicts

- If *A* and *B* are concurrent transactions, problems can occur if *A* and *B* want to read or write the same database object, say tuple *t*.
- There are four possibilities:
  - RR (Read Read): Reads cannot interfere with each other, so there is no problem in this case.
  - RW (Read Write)
  - WR (Write Read)
  - WW (Write Write)

RW Conflict

- *A* reads *t* and then *B* wants to write *t*.
- If *B* is allowed to perform its write, then the inconsistent analysis problem can arise.
  - As we saw in Fig. 16.4.
- Thus, we can say that inconsistent analysis is caused by a RW conflict.

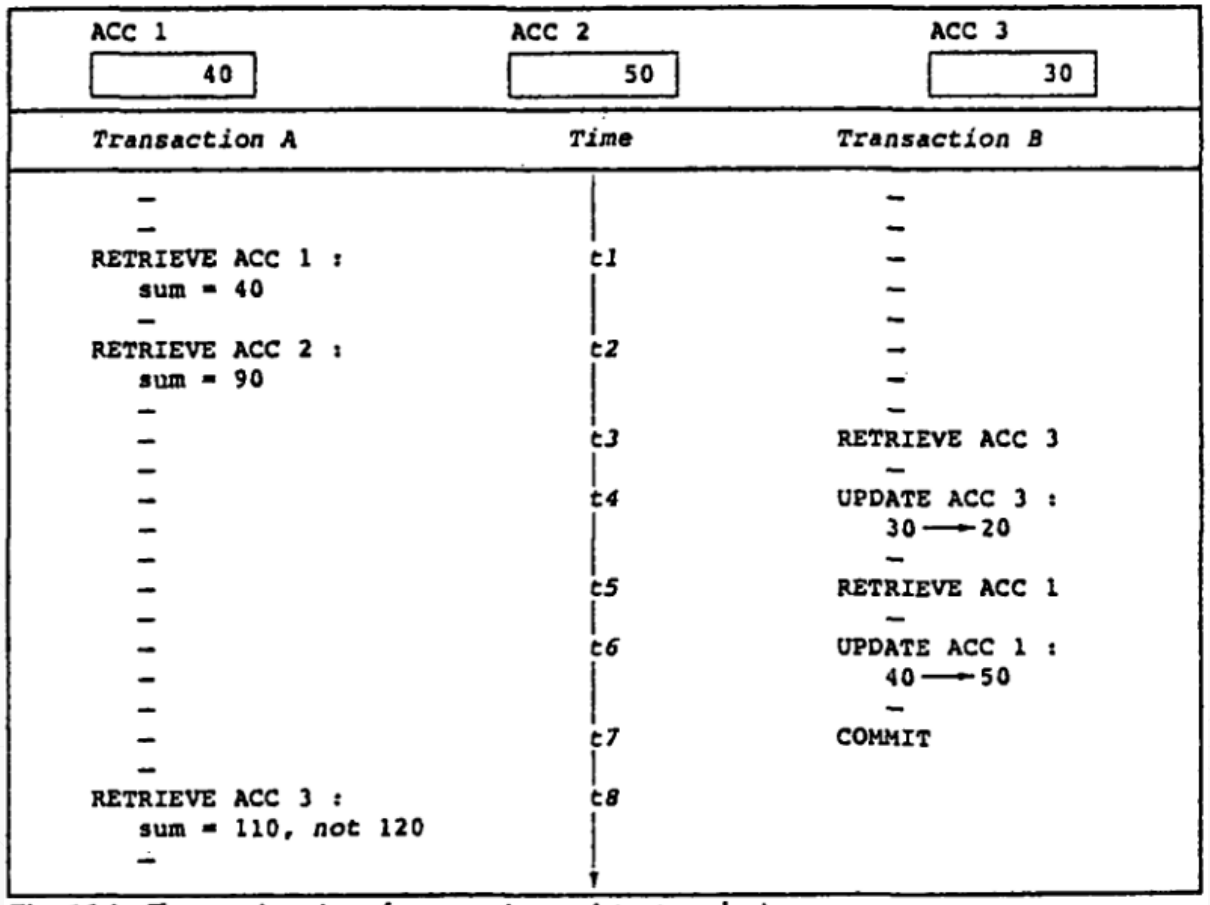


Fig. 16.4 Transaction *A* performs an inconsistent analysis

WR Conflict

- *B* writes *t* and then *A* wants to read *t*.
- If *A* is allowed to perform its read, then the uncommitted dependency problem can arise.
  - As we saw in Fig. 16.2.
- Thus, we can say that uncommitted dependencies are caused by WR conflicts.
- Note: *A*'s read, if it is allowed, is said to be a dirty read.

Transaction A	Time	Transaction B
–		–
–		–
–	t1	UPDATE t
–		–
RETRIEVE t	t2	–
–		–
–	t3	ROLLBACK
–		

Fig. 16.2 Transaction *A* becomes dependent on an uncommitted change at time *t*2

WW Conflict

- *A* writes *t* and then *B* wants to write *t*.
- If *B* is allowed to perform its write, then the lost update problem can arise.
  - As we saw in Fig. 16.1.
- Thus, we can say that lost updates are caused by WW conflicts.

Transaction A	Time	Transaction B
–		–
–		–
RETRIEVE t	t1	–
–		–
–	t2	RETRIEVE t
–		–
UPDATE t	t3	–
–		–
–	t4	UPDATE t
–		–

Fig. 16.1 Transaction *A* loses an update at time *t4*.

Locking

- The mentioned problems can all be solved by means of a concurrency control mechanism called locking.
- A transaction locks a portion of the database to prevent concurrency problems.
- Shared Lock (*S*)
  - Read lock
  - Locks out writes but allows other reads
- Exclusive Lock (*X*)
  - Write lock
  - Locks out all other transactions

Lock-Based Protocols

- Lock Compatibility Matrix:

	S	X
S	true	false
X	false	false

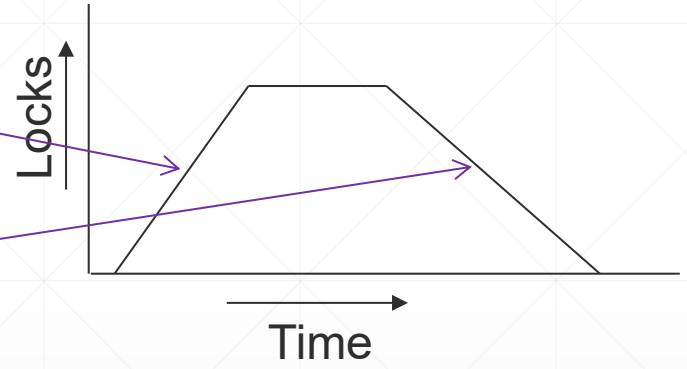
- A transaction may be granted a lock on an item if the requested lock is compatible with locks already held on the item by other transactions.
- Any number of transactions can hold shared locks on an item.
- But if any transaction holds an exclusive on the item no other transaction may hold any lock on the item.

Locking Protocol

- A locking protocol is a set of rules that state when a transaction may lock and unlock each of the data items in the database.
- **The Two-Phase Locking Protocol**

Two-Phase Locking Protocol

- Phase 1: **Growing Phase**
  - Transaction may obtain locks
  - Transaction may not release locks
- Phase 2: **Shrinking Phase**
  - Transaction may release locks
  - Transaction may not obtain locks



Example: Two-Phase Protocol

- Consider the transactions T_{34} and T_{35}:
  - Add lock and unlock instructions to transactions T_{34} and T_{35}, so that they observe the two-phase locking protocol.
    - A transaction requests a shared lock on data item Q by executing the lock-S(Q) instruction.
    - A transaction requests an exclusive lock on data item Q by executing the lock-X(Q) instruction.
    - A transaction can unlock a data item Q by the unlock(Q) instruction.

T_{34}: read(A);
read(B);
if $A = 0$ **then** $B := B + 1$;
write(B).

T_{35}: read(B);
read(A);
if $B = 0$ **then** $A := A + 1$;
write(A).

Example: Two-Phase Protocol (Cont.)

- Lock and unlock instructions:

T_{34}: read(A);
read(B);
if $A = 0$ then $B := B + 1$;
write(B).

T_{35}: read(B);
read(A);
if $B = 0$ then $A := A + 1$;
write(A).

T_{34}:	T_{35}:
lock-S(A)	lock-S(B)
read(A)	read(B)
lock-X(B)	lock-X(A)
read(B)	read(A)
if $A = 0$	if $B = 0$
then $B := B + 1$	then $A := A + 1$
write(B)	write(A)
unlock(A)	unlock(B)
unlock(B)	unlock(A)

Reading Uncommitted Data

- Releasing locks early (even with two-phase locking) is that transactions will be able to read uncommitted data.
- If T_1 eventually commits, then there is no problem.
  - But suppose that T_1 does a rollback.
    - Then T_2 will also have to rollback, because its execution is based on changes that no longer exist. And if T_2 rolls back, this could cause still other transactions to roll back.
  - This phenomenon is known as **cascading rollback**.

T_1	T_2
…	
	…
write(A)	
unlock(A)	
	lock-S(A)
	read(A)
ROLLBACK	

Extensions to Basic Two-Phase Locking Protocol

- **Two-phase locking:** Allows a transaction to lock a new data item only if that transaction has not yet unlocked any data item.
  - Not free from cascading rollbacks
- **Strict two-phase locking:** A transaction must hold all its exclusive locks till it commits/roll backs.
  - Ensures recoverability of freedom from cascading rollback.
- **Rigorous two-phase locking**: A transaction must hold *all* locks till commit/rollback.
  - Ensures recoverability of freedom from cascading rollback.
  - Transactions can be serialized in the order in which they commit.

Transaction A	Time	Transaction B
–		–
–		–
RETRIEVE t	t1	–
–		–
–	t2	RETRIEVE t
–		–
UPDATE t	t3	–
–		–
–	t4	UPDATE t
–		–

Fig. 16.1 Transaction *A* loses an update at time *t4* .

The Lost Update Problem

A lost update occurs when a second transaction reads the state of the database prior to the first one writing a change, and then stomps on the first one's change with its own update.

The Lost Update Problem - Revisited

Transaction A	Time	Transaction B
–		–
–		–
RETRIEVE t	t1	–
(acquire S lock on t)		–
–		–
–	t2	RETRIEVE t
–		(acquire S lock on t)
–		–
UPDATE t	t3	–
(request X lock on t)		–
wait		–
wait	t4	UPDATE t
wait		(request X lock on t)
wait		wait
wait		wait
wait		wait

Fig. 16.6 No update is lost, but deadlock occurs at time t4

Transaction A	Time	Transaction B
–		–
–		–
–	t1	UPDATE t
–		–
RETRIEVE t	t2	–
–		–
–	t3	ROLLBACK
–		

Fig. 16.2 Transaction *A* becomes dependent on an uncommitted change at time *t2*

The Uncommitted Dependency Problem

An uncommitted dependency occurs when a second transaction relies on a change which has not yet been committed, which is rolled back after the second transaction has begun.

The Uncommitted Dependency Problem - Revisited

Transaction A	Time	Transaction B
–		–
–		–
–	t1	UPDATE t
–		(acquire X lock on t)
–		–
RETRIEVE t	t2	–
(request S lock on t)		–
wait		–
wait	t3	COMMIT / ROLLBACK
wait		(release X lock on t)
resume : RETRIEVE t	t4	
(acquire S lock on t)		
–		

Fig. 16.7 Transaction *A* is prevented from seeing an uncommitted change at time *t2*

The Inconsistent Analysis Problem

An inconsistent analysis occurs when totals are calculated during interleaved updates.

ACC 1	ACC 2	ACC 3
40	50	30

Transaction A	*Time*	*Transaction B*
–		–
–		–
RETRIEVE ACC 1 :	*t1*	–
sum = 40		–
–		–
RETRIEVE ACC 2 :	*t2*	–
sum = 90		–
–		–
–	*t3*	RETRIEVE ACC 3
–		–
–	*t4*	UPDATE ACC 3 :
–		30 ⟶ 20
–		–
–	*t5*	RETRIEVE ACC 1
–		–
–	*t6*	UPDATE ACC 1 :
–		40 ⟶ 50
–		–
–	*t7*	COMMIT
–		
RETRIEVE ACC 3 :	*t8*	
sum = 110, *not* 120		
–		

Fig. 16.4 Transaction *A* performs an inconsistent analysis

The Inconsistent Analysis Problem - Revisited

ACC 1	ACC 2	ACC 3
40	50	30

Transaction A	Time	Transaction B
–		–
–		–
RETRIEVE ACC 1 :	t1	–
(acquire S lock on ACC 1)		–
sum = 40		–
–		–
RETRIEVE ACC 2 :	t2	–
(acquire S lock on ACC 2)		–
sum = 90		–
–		–
–	t3	RETRIEVE ACC 3
–		(acquire S lock on ACC 3)
–		–
–	t4	UPDATE ACC 3
–		(acquire X lock on ACC 3)
–		30 → 20
–		–
–	t5	RETRIEVE ACC 1
–		(acquire S lock on ACC 1)
–		–
–	t6	UPDATE ACC 1
–		(request X lock on ACC 1)
–		wait
RETRIEVE ACC 3 :	t7	wait
(request S lock on ACC 3)		wait
wait		wait
wait		wait

Fig. 16.9 Inconsistent analysis is prevented, but deadlock occurs at time *t7*

Deadlock

- Consider the partial schedule:
- Neither T_3 nor T_4 can make progress — executing **lock-S**(B) causes T_4 to wait for T_3 to release its lock on B, while executing **lock-X**(A) causes T_3 to wait for T_4 to release its lock on A.
- Such a situation is called a **deadlock**.
  - To handle a deadlock one of T_3 or T_4 must be rolled back and its locks released.
- The potential for deadlock exists in most locking protocols.

T_3	T_4
lock-X(B)	
read(B)	
$B := B - 50$	
write(B)	
	lock-S(A)
	read(A)
	lock-S(B)
lock-X(A)	

Deadlock Handling

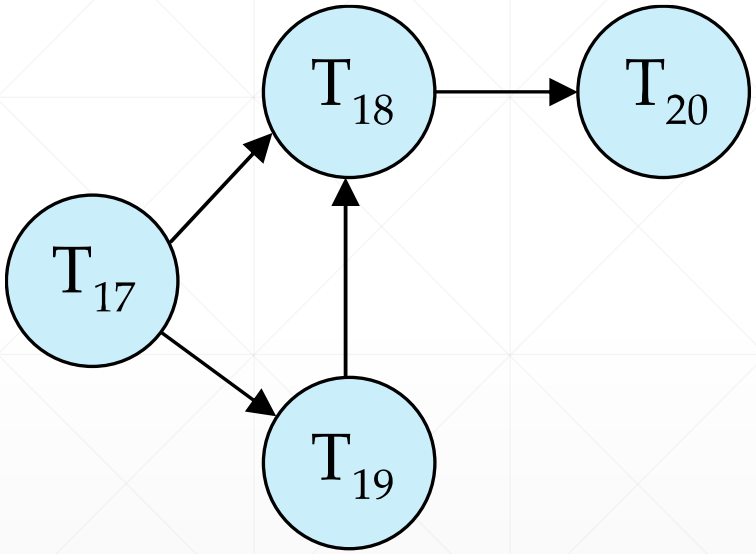
- Various locking protocols do not guard against deadlocks.
- System is **deadlocked** if there is a set of transactions such that every transaction in the set is waiting for another transaction in the set.

T_3	T_4
lock-X(B)	
read(B)	
$B := B - 50$	
write(B)	
	lock-S(A)
	read(A)
	lock-S(B)
lock-X(A)	

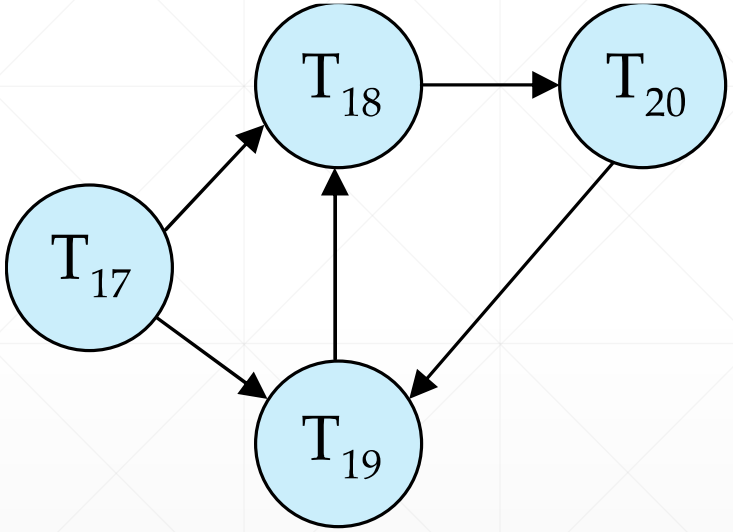
Deadlock Detection

- If deadlocks are not prevented, the system must deal with them by using a deadlock detection and recovery scheme.
- **Wait-for graph**
  - *Vertices:* transactions
  - *Edge from* $T_i \rightarrow T_j$. : if T_i is waiting for a lock held in conflicting mode by T_j
- The system is in a deadlock state if and only if the wait-for graph has a cycle.
- Invoke a deadlock-detection algorithm periodically to look for cycles.

Deadlock Detection (Cont.)



Wait-for graph without a cycle



Wait-for graph with a cycle

Deadlock!

Deadlock Prevention Strategies

- The locking protocol may be modified to avoid deadlock by using Wait-Die Scheme and Wound-Wait Scheme
- **Wait-Die**: Transaction 2 waits if it is older than 1; otherwise it dies (rolls back).
- **Wound-Wait:** Transaction 2 wounds 1 if it is older; that is, it rolls it back; otherwise it waits.
- Net effect: oldest transaction wins
- In both schemes, a rolled back transaction is restarted with its original timestamp.
  - Ensures that older transactions have precedence over newer ones.

Example: Wait-Die

T1	T2
lock-S(A)	
read(A)	
	lock-S(B)
	read(B)
lock-X(B)	
	lock-X(A)

- T1 is older so it is allowed to wait.
- T2 is younger so it is roll backed.
  - Its locks are released.
  - Allows T1 to proceed.

Example: Wound-Wait

T1	T2
lock-S(A)	
read(A)	
	lock-S(B)
	read(B)
lock-X(B)	

- T1 is older so T2 is roll backed and that allows to T1 proceed.

Deadlock Prevention Strategies (Cont.)

Wait-Die Scheme — non-preemptive

- Older transaction may wait for younger one to release data item.
- Younger transactions never wait for older ones; they are rolled back instead.
- A transaction may die several times before acquiring a lock.

Wound-Wait Scheme — preemptive

- Older transaction *wounds* (forces rollback) of younger transaction instead of waiting for it.
- Younger transactions may wait for older ones.
- Fewer rollbacks than *wait-die* scheme.

Deadlock Prevention Strategies (Cont.)

- **Timeout-Based Schemes**:
  - A transaction waits for a lock only for a specified amount of time. After that, the wait times out and the transaction is rolled back.
  - Ensures that deadlocks get resolved by timeout if they occur.
  - Simple to implement.
  - But may roll back transaction unnecessarily in absence of deadlock
    - Difficult to determine good value of the timeout interval.

Serializability

- A ***serial schedule*** is a schedule in which the histories are not interleaved.
- A ***serializable schedule*** is a schedule that produces the same result as some serial schedule.
- Responsible for the correct execution of concurrent transactions.
- What is “correct” execution?
  - Serializable Schedule!
- A schedule is ***correct*** if and only if it is ***serializable***.

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
	$B := B + temp$
	write (B)
	commit

Schedule 1

- Let T_1 transfer \$50 from A to B, and T_2 transfer 10% of the balance from A to B.
- A serial schedule in which T_1 is followed by T_2

T_1	T_2
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
	$B := B + temp$
	write (B)
	commit
read (A)	
$A := A - 50$	
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	

Schedule 2

A serial schedule where T_2 is followed by T_1

T_1	T_2
read (A)	
$A := A - 50$	
write (A)	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
read (B)	
$B := B + 50$	
write (B)	
commit	
	read (B)
	$B := B + temp$
	write (B)
	commit

Schedule 3

- Let T_1 and T_2 be the transactions defined previously.
- The following schedule is not a serial schedule, but it is *equivalent* to Schedule 1
- In Schedules 1, 2 and 3, the sum A + B is preserved.

T_1	T_2
read (A)	
$A := A - 50$	
	read (A)
	$temp := A * 0.1$
	$A := A - temp$
	write (A)
	read (B)
write (A)	
read (B)	
$B := B + 50$	
write (B)	
commit	
	$B := B + temp$
	write (B)
	commit

Schedule 4

The following concurrent schedule does not preserve the value of (A + B).

Serializability

- An interleaved execution is considered correct if and only if it is serializable.
- A set of transactions is serializable if and only if it is guaranteed to produce the same result as when each transaction is completed prior to the following one being started.
- If all transactions obey two phase locking, then all possible interleaved schedules are serializable.

Transaction Isolation Levels

- Serializable transactions can run very slowly.
- Clients can specify how much isolation they want their transaction to have (+ problems)
- Apply to only to data reading
- All transactions, regardless of their isolation level, behave correctly with respect to writing data

Transaction Isolation Levels

- **Read uncommitted:**
  - Means no isolation at all
  - Even uncommitted values may be read
- **Read committed:**
  - Forbids a transaction from accessing uncommitted values
  - Successive reads of record may return different (but committed) values
- **Repeatable read:**
  - Extends read committed so that reads are always repeatable
- **Serializable:**
  - Ensures serializable execution

Dirty Read, Nonrepeatable Read and Phantom

- **Dirty read:** A transaction reads values written by another transaction that hasn't committed yet.
- **Nonrepeatable read:** A transaction reads the same object twice during execution and finds a different value the second time, although the transaction has not changed the value in the meantime.
- **Phantom read:** A transaction re-executes a query returning a set of rows that satisfy a search condition and finds that the set of rows satisfying the condition has changed as a result of another recently committed transaction.

Transaction Isolation Levels and Problems

ISOLATION LEVEL	DIRTY READ	NONREPEATABLE READ	PHANTOM READ
READ UNCOMMITTED	Y	Y	Y
READ COMMITTED	N	Y	Y
REPEATABLE READ	N	N	Y
SERIALIZABLE	N	N	N

Transaction Isolation Levels and Shared Locks

- **Read uncommitted:**
  - Means no isolation at all
  - Even uncommitted records may be read
- **Read committed:**
  - Forbids a transaction from accessing uncommitted values
  - Successive reads of record may return different (but committed) values
- **Repeatable read:**
  - Extends read committed so that reads are always repeatable
- **Serializable:**
  - Ensures serializable execution

Does not even use shared locks

Very restrictive shared locking

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